AP CALCULUS AB - UNIT 4

Guided Notes 4.2: Straight Line Motion

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we relate position, velocity, speed, and acceleration using derivatives?

- Relate position, velocity, speed, and acceleration using derivatives.
- Determine direction of motion, rest times, and changes in speed.
- Analyze motion from formulas, graphs, and tables.
- Calculate displacement and total distance over intervals using turning times.

Key Knowledge, Vocabulary, and Notation

Position	$s(t)$ locates a particle on a line; units are distance units.
Velocity	$v(t) = s'(t)$; its sign determines direction.
Speed	$\lvert v(t)\rvert$; speed is never negative.
Acceleration	$a(t) = v'(t) = s''(t)$; it measures change in velocity.
Speeding up	Velocity and acceleration have the same sign.
Slowing down	Velocity and acceleration have opposite signs.
Displacement	$s(b) - s(a)$; total distance adds absolute position changes across direction reversals.

Concept Development and Strategy

1. Find velocity and acceleration before solving sign questions.
2. A particle moves right when $v > 0$, left when $v < 0$, and is at rest when $v = 0$.
3. Use a sign chart for both v and a; do not infer speeding up from $a > 0$ alone.
4. To compute total distance without integration, locate all times where velocity changes sign and add absolute changes in position.
5. A zero velocity can be a stop without a direction reversal if the sign of velocity does not change.

Shared Representation / Data

t	0	1	2	3	4	5	6
$s(t)$	4	7	8	7	4	−1	−8
$v(t)$	4	2	0	−2	−4	−6	−8
$a(t)$	−2	−2	−2	−2	−2	−2	−2

Worked Example: Direction and rest

For $s(t) = t^3 - 6t^2 + 9t$, find the velocity and the times the particle is at rest.

Answer and Reasoning

$v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$. The particle is at rest at $t = 1$ and $t = 3$.

Worked Example: Speeding up

For $v(t) = t^2 - 4t + 3$ and $a(t) = 2t - 4$, determine where the particle speeds up on $0 < t < 5$.

Answer and Reasoning

Velocity is positive on $(0,1) \cup (3,5)$ and negative on $(1,3)$; acceleration is negative on $(0,2)$ and positive on $(2,5)$. Same signs occur on $(1,2) \cup (3,5)$.

Worked Example: Total distance

A particle has $s(t) = t^3 - 3t^2$ for $0 \le t \le 4$. Find total distance.

Answer and Reasoning

$v = 3t(t-2)$ changes sign at $t = 2$. Positions: $s(0) = 0$, $s(2) = -4$, $s(4) = 16$. Distance $= |-4-0| + |16-(-4)| = 24$.

Worked Example: Motion from a table

Using the shared table, describe motion at $t = 1$ and $t = 4$.

Answer and Reasoning

At $t = 1$, $v = 2 > 0$ and $a = -2 < 0$: moving right and slowing down. At $t = 4$, $v = -4$ and $a = -2$: moving left and speeding up.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. If $s(t) = 5t^2 - 3t + 1$, find $v(t)$ and $a(t)$.

2. If $v(4) = -7$, state the direction and speed at $t = 4$.

3. For $s(t) = t^3 - 9t$, find all rest times for $t \geq 0$.

4. For $v(t) = (t-2)(t-5)$, determine intervals of rightward motion.

5. For $s(t) = t^3 - 6t^2 + 9t$, determine intervals of right and left motion for $t > 0$.

6. For $v(t) = t^3 - 3t^2 - 9t + 27$, determine all intervals on which speed increases.

7. Construct a differentiable velocity function that is zero at $t = 1$ but does not reverse direction there.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. A particle is slowing down when

(A) $v > 0, a > 0$

(B) $v < 0, a < 0$

(C) v and a have opposite signs

(D) $a = 0$

2. If $s'(4) = -3$, then at $t = 4$ the particle

(A) moves right at speed 3

(B) moves left at speed 3

(C) has acceleration -3

(D) is at rest

3. For $s(t) = t^3 - 4t$, the velocity at $t = 1.7$ is closest to

(A) 1.33

(B) 4.67

(C) 6.27

(D) 8.67

Common Errors and AP Precision

- State the quantity, sign, input value, and units in every contextual interpretation.
- In related rates, differentiate before substituting values. In linearization, identify the base point. For L'Hopital's Rule, verify $0/0$ or ∞/∞ first.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Suppose $s'(t) = |t - 2|$. Is s twice differentiable at $t = 2$? Explain.